

## AICR · OPEN PROBLEM

# Groth16 Is the Limit in the Pure Generic Group Model

*Two lower-bound conjectures on pairing-based SNARGs*

**Status.** Statement: AI-written, not yet formalized.

**Category.** Proof Systems & Succinct Arguments.

**Conjecture (informal).** *In the pure generic bilinear group model — no random oracle or any other channel from group elements to bits; the prover can only apply group operations to reference-string elements; the verifier checks pairing product equations fixed by the statement — every sound, nontrivially complete pairing-based SNARG for a hard language has at least three group elements. Groth16 is proof-size optimal.*

## 1 The setting

In a pairing-based succinct non-interactive argument (SNARG) the prover publishes a few group elements and the verifier checks pairing equations; no shorter proof systems are known. Groth’s argument for arithmetic circuit satisfiability [Gro16] — Groth16 — needs only three: two elements of  $\mathbb{G}_1$  and one of  $\mathbb{G}_2$ , verified by a single pairing product equation, knowledge-sound against generic adversaries. The same paper rules out single-element proofs and closes with the problem taken up here: whether the bound extends to two elements, “which would prove optimality of our 3 element construction” [Gro16]. Section 2 formalizes the statement above: in the *purist* reading of the model, where Groth16’s own security proof lives, the answer is negative.

**Notation.**  $(p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, [1]_1, [1]_2)$  is an asymmetric (Type III) bilinear group of prime order  $p$ : no efficient homomorphism exists in either direction between  $\mathbb{G}_1$  and  $\mathbb{G}_2$ . Brackets denote discrete logarithms:  $[a]_i \in \mathbb{G}_i$  for  $a \in \mathbb{F} := \mathbb{Z}_p$ , groups are written additively,  $[a]_1 \cdot [b]_2 = [ab]_T$ , and linear algebra lifts entrywise,

$M[a]_i = [Ma]_i$ ;  $\langle \cdot, \cdot \rangle$  is the inner product and  $(a; b)$  vertical concatenation. Everything below is publicly verifiable; nothing assumes zero knowledge — impossibilities are for plain soundness, a fortiori for stronger notions.

**The purist model.** In Shoup’s generic group model [Sho97] elements are random strings an adversary may compare, hash, or take apart bit by bit; in Maurer’s [Mau05] they live behind handles, touchable only through the group oracles — no channel from group elements back to bits. Groth’s construction and lower bounds [Gro16] concern provers that pick matrices and apply them to the CRS “in the exponent”, and verifiers that check pairing product equations with proof-independent coefficients. We call such a scheme *purist* when the prover is generic in Maurer’s sense with no auxiliary oracles (the name *pure GGM* follows [ADY25]): in particular, no random oracle. Groth flags the CRS-side analogue (a reference string  $G^b$  encoding bits  $b$ ) [Gro16]; fact (F3b) shows that restoring the prover-side channel collapses the proof to two elements. Definition 4 formalizes the model.

**What is known.** Four facts fix the landscape.

- (F1) *Three elements suffice.* Groth16 is, at its core, a split non-interactive linear proof (Definition 2) of dimension  $(2, 1)$  for quadratic arithmetic programs: perfectly complete, perfectly zero-knowledge, statistically knowledge-sound against affine provers (Definition 3) [Gro16], in the formalism of [BCIOP13]. Compiled:  $2 \mathbb{G}_1 + 1 \mathbb{G}_2$  (1536 bits over BLS12-381), one pairing product equation.
- (F2) *One element is impossible — robustly.* NILPs with a degree-1 decision procedure do not exist for relation generators with hard decisional problems (Definition 1), so a purist proof needs elements of both source groups,  $k_1, k_2 \geq 1$ : at least two elements [Gro16, Thms. 3–4]. Assuming one-way functions, Arnon, Dujmovic and Yogev [ADY25] extend the one-element impossibility to *arbitrary* Maurer-generic verifiers (restriction (iii) of Definition 4 dropped) and, for non-adaptive verifier queries and oracle-free setups, to the GGM + ROM.
- (F3) *Every purist restriction is load-bearing.* (a) *Symmetry.* In Type I groups ( $\mathbb{G}_1 = \mathbb{G}_2$ ) two elements suffice: squaring gates force

$u_i = v_i$ , so  $B = A + \beta - \alpha$  and the verifier recomputes  $[B]$  itself, leaving  $(A, C)$  [Gro16, Sect. 3.1]. Two *field* elements likewise suffice at the bare NILP level, so no dimension count over  $\mathbb{F}$  can prove the conjectures: the split is the crux. (b) *Random oracles*. In the GGM + ROM — Maurer’s model plus an oracle taking group elements to bits — two elements suffice, both in  $\mathbb{G}_1$  (768 bits over BLS12-381): the prover of [ADY25] feeds its first proof element through the oracle to derive the verifier’s otherwise-unpredictable query — exactly the channel the purist model lacks — and the underlying PCP has accepting set of size 1, so no answer is sent. (Completeness becomes imperfect; negligible errors need  $\lambda \leq \log^c n$ , or Khot’s Unique Games Conjecture [Kho02], used for completeness, not soundness.) (c) *Non-group data*. Field elements in the proof buy fewer bits with more elements: Polymath,  $3\mathbb{G}_1 + 1\mathbb{F} \approx 1408$  bits [Lip24]; Pari,  $2\mathbb{G}_1 + 2\mathbb{F} \approx 1280$  bits [DMS24]. Element count and bit count thus part ways; the conjectures concern the former. (d) *Interaction; designated verifiers*. One round of interaction yields one-element proofs [BIOW20]; designated-verifier SNARGs reach one element plus  $O(\lambda)$  bits [ADI25]. Non-interactivity and public verifiability are load-bearing too.

- (F4) *Simulation extractability is settled*. Pairing-based simulation-extractable SNARKs need at least 3 elements and 2 verification equations, matched by [GM17]; Groth16 proofs are perfectly rerandomizable, so simulation extractability is strictly stronger and the bound does not transfer.

**The gap.** By F2 a two-element purist proof is exactly  $\pi = ([A]_1, [B]_2)$ , and in the exponent each verification equation reads

$$c_i \cdot AB + A \langle u_i, \sigma_2 \rangle + \langle v_i, \sigma_1 \rangle B + \sigma_1^\top M_i \sigma_2 = 0.$$

Every term is affine in the prover’s coefficients except the single bilinear monomial  $AB$  — and the sampling attacks behind [Gro16, Thms. 3–4] and the one-element bound of [ADY25] die on exactly that monomial (Remark 1). The no-ROM two-element question is [ADY25]’s first open problem; the conjectures assert the negative answer.

## 2 The conjectures

Notation as in Section 1;  $\lambda$  is the security parameter. Definitions follow [Gro16], restated for self-containment; the conjectures quantify over all relation generators.

**Definition 1 (relation generator; hard decisional problems [Gro16, Def. 5]).** A *relation generator*  $\mathcal{R}$  returns on  $1^\lambda$  a polynomial-time decidable binary relation  $R$  (with statements  $\phi$ , witnesses  $w$ , and language  $L_R := \{\phi : \exists w, (\phi, w) \in R\}$ ) together with auxiliary input  $z$ . It has *hard decisional problems* if there are polynomial-time samplers with  $\text{Yes}(R) \rightarrow (\phi, w) \in R$  and  $\text{No}(R) \rightarrow \phi \notin L_R$  (with overwhelming probability), such that for all polynomial-time distinguishers  $\mathcal{A}$ , over  $\phi_0 \leftarrow \text{No}(R)$ ,  $(\phi_1, w_1) \leftarrow \text{Yes}(R)$  and  $b \leftarrow \{0, 1\}$ ,

$$\Pr[\mathcal{A}(R, z, \phi_b) = b] \approx \frac{1}{2}.$$

Any NP-complete relation qualifies if one-way functions exist [Gro16].

**Definition 2 (split NILP of proof dimension  $(k_1, k_2)$  [Gro16, Sect. 2.5]).** A *split non-interactive linear proof* for  $\mathcal{R}$  is a triple  $(\text{Setup}, \text{Prove}, \text{Vfy})$  of the following form.  $\text{Setup}(R)$  outputs  $\sigma = (\sigma_1, \sigma_2) \in \mathbb{F}^{m_1} \times \mathbb{F}^{m_2}$ , each part containing 1 as an entry, and a simulation trapdoor  $\tau \in \mathbb{F}^n$ .  $\text{Prove}(R, \sigma, \phi, w)$  samples  $(\Pi_1, \Pi_2) \leftarrow \text{ProofMatrix}(R, \phi, w)$  with  $\Pi_j \in \mathbb{F}^{k_j \times m_j}$  and outputs  $\pi = (\pi_1, \pi_2) = (\Pi_1 \sigma_1, \Pi_2 \sigma_2)$ .  $\text{Vfy}(R, \sigma, \phi, \pi)$  derives deterministic tests  $(T_1, \dots, T_\eta) \leftarrow \text{Test}(R, \phi)$  with  $T_i \in \mathbb{F}^{(m_1+k_1) \times (m_2+k_2)}$ , and accepts if and only if

$$(\sigma_1; \pi_1)^\top T_i (\sigma_2; \pi_2) = 0 \quad \text{for all } i = 1, \dots, \eta.$$

The NILP has *completeness error*  $\sigma$  if honest proofs for  $(\phi, w) \in R$  verify with probability at least  $1 - \sigma$ ; Groth16's NILP is perfectly complete,  $\sigma = 0$ .

**Definition 3 (statistical soundness against affine prover strategies [Gro16, Def. 6]).** An *affine prover strategy* chooses  $(\phi, \Pi_1, \Pi_2)$  given only  $(R, z)$  — in particular, independently of  $\sigma$ . A split NILP is *statistically sound against affine prover strategies* if for all (unbounded) adversaries  $\mathcal{A}$ , over  $(R, z) \leftarrow \mathcal{R}(1^\lambda)$ ,  $(\sigma, \tau) \leftarrow \text{Setup}(R)$  and  $(\phi, \Pi_1, \Pi_2) \leftarrow \mathcal{A}(R, z)$ ,

$$\Pr \left[ \phi \notin L_R \wedge \text{Vfy}(R, \sigma, \phi, (\Pi_1 \sigma_1, \Pi_2 \sigma_2)) = 1 \right] \approx 0.$$

**Definition 4 (purist pairing-based argument).** A *purist pairing-based non-interactive argument* for  $\mathcal{R}$  is a publicly verifiable argument system in which (i) the reference string is a group description plus group elements only,  $\sigma = ([\sigma_1]_1, [\sigma_2]_2, [\sigma_T]_T)$ ; (ii) the prover is generic in Maurer’s sense [Mau05] — it touches group elements only through the group, pairing and equality oracles, and never sees an encoding — so its proof is

$$\pi = ([\Pi_1 \sigma_1]_1, [\Pi_2 \sigma_2]_2, [\Pi_T \sigma_T]_T)$$

with matrices depending only on  $R, \phi$  and the observed equality pattern; there are no auxiliary oracles: no random oracle, no channel from group elements to bits; and (iii) from  $\phi$  alone the verifier derives matrices  $T_i$  and vectors  $t_i$  — pairing product equations with proof-independent coefficients — and accepts if and only if

$$[\sigma_1; \pi_1]_1^\top T_i [\sigma_2; \pi_2]_2 = [t_i \cdot (\sigma_T; \pi_T)]_T \quad \text{for all } i.$$

The *proof size* is  $k_1 + k_2 + k_T$ ; completeness error is as for split NILPs; *soundness*: every polynomial-query purist prover outputs an accepted  $(\phi, \pi)$  with  $\phi \notin L_R$  with negligible probability. Groth16 is a sound, perfectly complete purist argument of size  $(2, 1, 0)$  [Gro16].

**Conjecture 1 (no two-element split NILPs).** *There are no split NILPs with proof dimension  $k_1 = k_2 = 1$  and completeness error at most  $1 - 1/\text{poly}(\lambda)$  for relation generators with hard deci-*

*sional problems. Concretely: any such candidate (Definition 2) yields either an affine prover strategy breaking soundness (Definition 3), or a polynomial-time algorithm distinguishing the Yes- and No-samplers of Definition 1 with noticeable advantage.*

**Conjecture 2 (Groth16 optimality in the pure GGM).** *Every sound purist pairing-based non-interactive argument (Definition 4) with completeness error at most  $1 - 1/\text{poly}(\lambda)$  for a relation generator with hard decisional problems has proof size at least three. Equivalently: two-group-element purist SNARGs do not exist, and Groth16 is proof-size optimal in the pure GGM.*

An affine strategy is a purist prover making no equality tests, so Conjecture 1 implies Conjecture 2; the converse could fail only via reference strings that are not disclosure-free [Gro16, Def. 4, Lemma 1]. A (1,1) split NILP for a hard relation generator refutes Conjecture 1 — and Conjecture 2 if disclosure-free; no zero knowledge is required. Dropping (iii) as well — arbitrary Maurer-generic verifiers, as in [ADY25] — gives a stronger form we also conjecture.

**Remark 1 (why the model should have no room for two).** Groth16’s third element  $C$  absorbs the cross terms  $As + rB - rs\delta$  from randomizing  $A$  and  $B$ , while  $\alpha\beta$ ,  $\gamma$  and  $\delta$  separate statement-from witness-consistency in one equation. The ROM route (F3b) is hide-then-reveal — commit to the first element *before* the test is drawn — while the purist  $T_i$  are fixed by  $\phi$ . The one-element bounds run on uniqueness: a proof-dependent zero-test is affine in the lone element and pins a single accepting value, which yes-instance sampling learns [Gro16, ADY25]; with  $k_1 = k_2 = 1$  it pins only a Möbius curve of pairs  $(A, B)$ , which the sampling attack no longer isolates. The conjecture:  $AB$  still cannot both couple the sides’ witness dependence and resist sampling.

**Remark 2 (a half-remembered lead, recorded with low confidence).** The author recalls, but cannot reconstruct, a sketch by which a two-element scheme would contradict a complexity conjecture — possibly about *unique witnesses*, à la Valiant–Vazirani [VV86]; a lead, not a claim. Two facts orbit it: the all-parameter construction of [ADY25] assumes the Unique Games Conjecture [Kho02] (a

“unique”-flavoured conjecture on the *constructive* side; the memory may be that, reversed), and by Remark 1 a (1,1) scheme flirts with unique accepting proofs; making this precise — or refuting it — is part of the problem.

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